



6.4 The Salivary Glands

6.4.0 Embryology and Anatomy of the Salivary Glands

◆ Key Features

- The salivary glands begin to form at 6 to 9 weeks gestation.
- The major salivary glands arise from ectodermal tissue.
- Salivary glands are composed of acini and ducts.
- The parotid gland is the largest of the salivary glands, and the submandibular glands are the second largest, followed by the sublingual glands.

◆ Embryology

The embryologic development of the salivary glands is the result of complex interaction between the oral epithelium and the underlying mesenchyme, with a contribution also from neural crest. The salivary glands share a common embryogenesis in that they develop from growths of oral epithelium into the underlying mesenchyme. The acini (secretory units) and the ductal system of each gland will eventually arise from these epithelial outgrowths, which are of ectodermal origin for the parotid glands, submandibular glands, and sublingual glands, and are of mixed ectodermal and endodermal origin for the minor salivary glands. The epithelial cells carry the information for the type of salivary secretions that will be produced by each future gland, while the mesenchymal cells contain the information for the branching pattern that eventually will be the morphologic signature of these glands. The stroma, which comprises the capsule of each gland as well as the septa that divide the gland into lobes and lobules, will develop from cranial neural crest cells.

The parotid anlagen appear first, between the fourth and sixth embryonic weeks, as solid epithelial placodes in the developing cheeks. The placodes for the submandibular glands appear later in the sixth embryonic week. During the seventh to eighth embryonic weeks, the sublingual gland anlagen arise from multiple epithelial placodes, lateral to the submandibular glands, and finally the minor salivary glands develop late in the 12th fetal week.

◆ Anatomy

Parotid Glands

The paired parotid glands are the largest of the major salivary glands and typically weigh 15 to 30 g. Located in the preauricular region and along the posterior surface of the mandible, each parotid gland is divided by the facial nerve into a superficial lobe and a deep lobe. The *superficial lobe*, overlying the lateral surface of the masseter, is defined as the part of the gland lateral to the facial nerve. The *deep lobe* is medial to the facial nerve and located between the mastoid process of the temporal bone and the ramus of the mandible. The parotid gland is bounded superiorly by the zygomatic arch. Inferiorly, the tail of the parotid gland abuts the anteromedial margin of the sternocleidomastoid muscle. This tail of the parotid gland extends posteriorly over the superior border of the sternocleidomastoid muscle toward the mastoid tip. The deep lobe of the parotid lies within the parapharyngeal space. The parotid duct (Stensen) secretes a serous saliva into the oral cavity. From the anterior border of the gland, it travels parallel to the zygoma, in an anterior direction across the masseter muscle. It then turns sharply to pierce the buccinator muscle and enters the oral cavity opposite the second upper molar tooth.

The submandibular gland is the second largest major salivary gland and weighs 7 to 16 g. The gland is situated in the *submandibular triangle*, its superior boundary is formed by the inferior edge of the mandible, and its inferior boundaries are formed by the anterior and posterior bellies of the digastric muscle. The *submandibular triangle* also contains the submandibular lymph nodes, facial artery and vein, mylohyoid muscle, and the lingual, hypoglossal, and mylohyoid nerves. The marginal mandibular branch of the facial nerve courses along the capsule of the gland laterally. Most of the submandibular gland lies posterolateral to the mylohyoid muscle. Often, smaller, tongue-like projections of the gland follow the duct, as it ascends toward the oral cavity, deep to the mylohyoid muscle (**Fig. 6.18**).

The submandibular gland has both mucous and serous cells that empty into ductules, which in turn empty into the *submandibular duct*. The duct exits anteriorly from the sublingual aspect of the gland, coursing deep to the lingual nerve and medial to the sublingual gland. It eventually forms the Wharton duct between the hyoglossus and mylohyoid muscles on the genioglossus muscle. The Wharton duct, the main excretory duct of the submandibular gland, is approximately 4 to 5 cm long, running superior to the hypoglossal nerve while inferior to the lingual nerve. It empties lateral to the lingual frenulum through a papilla in the floor of the mouth behind the lower incisor tooth. The openings for the sublingual gland, or the *sublingual caruncles*, are located near the midline of the sublingual fold in the ventral tongue.

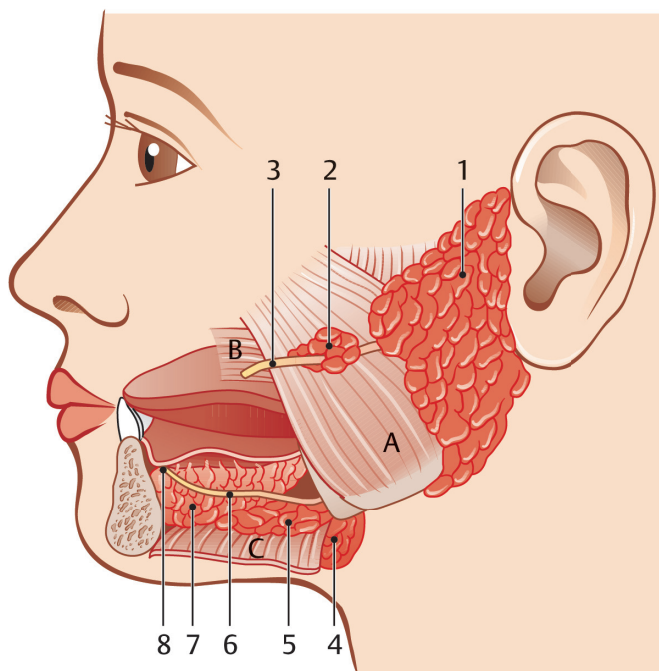


Fig. 6.18 The major salivary glands. Parotid gland (1) with small accessory gland (2) and Stensen duct (3). Submandibular gland (4) with uncinus process (5) and submandibular (Wharton) duct (6). Sublingual gland (7) with sublingual caruncle (8). A, masseter muscle; B, buccinator muscle; C, mylohyoid muscle. (Used with permission from Behrbohm H et al. *Ear, Nose, and Throat Diseases: With Head and Neck Surgery*. 3rd ed. Stuttgart/New York: Thieme; 2009:413.)

The sublingual glands, the smallest of the major salivary glands, typically weigh 2 to 4 g. Consisting mainly of mucous acinar cells, they lie as a flat structure in a submucosal plane within the anterior floor of the mouth, superior to the mylohyoid muscle and deep to the sublingual folds opposite the lingual frenulum. Several ducts from the superior portion of the sublingual gland either secrete directly into the floor of mouth or empty into the *Bartholin duct*, which then drains into the Wharton duct.

Approximately 1,000 minor salivary glands, ranging in size from 1 to 5 mm, line the oral cavity and oropharynx. The greatest number of these glands is in the lips, tongue, buccal mucosa, and palate. They can also be found along the tonsils, supraglottis, and paranasal sinuses. Each gland has a single duct, which secretes directly into the oral cavity; saliva can be serous, mucous, or mixed.

Arteries

The blood supply to the parotid gland is from branches of the *external carotid artery*. The external carotid courses superiorly from the carotid bifurcation and parallel to the mandible under the posterior belly of the

digastric muscle. The artery travels medial to the parotid gland and splits into two terminal branches. The *superficial temporal artery* runs superiorly from the superior portion of the parotid gland to the scalp within the superior pretragal region. The *maxillary artery* leaves the medial portion of the parotid to supply the infratemporal and pterygopalatine fossae.

The *transverse facial artery* branches off the superficial temporal artery and runs anteriorly between the zygoma and parotid duct to supply the parotid gland and duct. The submandibular as well as the sublingual glands are supplied by the *submental* and *sublingual arteries*, branches of the lingual and facial arteries. The *facial artery* is the main arterial blood supply of the submandibular gland.

Veins

The retromandibular vein, formed by the union of the maxillary vein and the superficial temporal vein, runs through the parotid gland just deep to the facial nerve to join the external jugular vein. This vein is often crossed by the facial nerve branches.

Lymphatics

The parotid harbors two nodal layers, both of which drain into the superficial and deep cervical lymph systems. Approximately 90% of the nodes are located in the superficial layer between the glandular tissue and its capsule. The parotid gland, external auditory canal, pinna, scalp, eyelids, and lacrimal glands are all drained by these superficial nodes. The deep layer of nodes drains the gland, external auditory canal, middle ear, nasopharynx, and soft palate.

In contrast to the parotid gland, the lymph nodes draining the submandibular gland are located between the gland and its fascia but are not embedded in the glandular tissue. They lie in close approximation to the facial artery and vein at the superior aspect of the gland and empty into the deep cervical and jugular chains.

Nerves

Salivary flow is regulated predominantly by the autonomic nervous system. Although both sympathetic and parasympathetic stimulation produces saliva, the parasympathetic system is dominant. The glossopharyngeal nerve, cranial nerve (CN) IX, provides visceral secretory innervation to the parotid gland. The nerve carries preganglionic parasympathetic fibers from the inferior salivatory nucleus in the medulla oblongata through the jugular foramen. Distal to the inferior ganglion, the *tympanic nerve*, a branch of the glossopharyngeal nerve, enters the skull through the inferior tympanic canaliculus and into the middle ear to form the *tympanic plexus*. The preganglionic fibers course along as the lesser petrosal nerve into the middle cranial fossa and out the foramen ovale to synapse in the *otic ganglion*. Postganglionic parasympathetic fibers exit the otic ganglion beneath the mandibular nerve to join the auriculotemporal nerve in the infratemporal fossa. These fibers innervate the parotid gland for the secretion of saliva. Postganglionic sympathetic fibers innervate salivary glands, sweat glands,

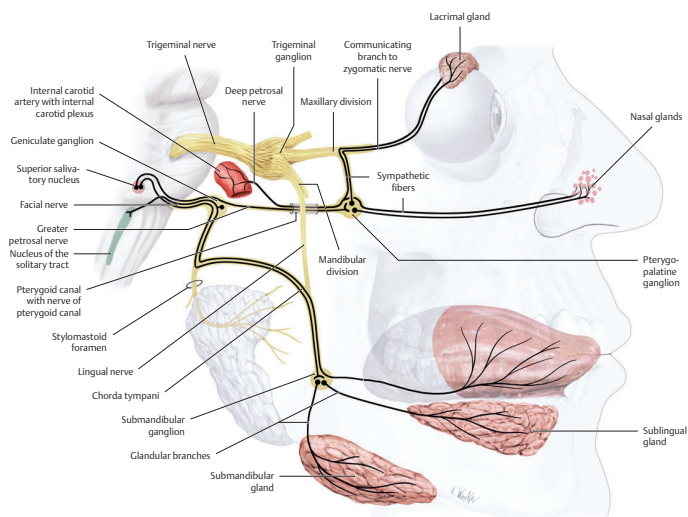


Fig. 6.19 The nervus intermedius supplies taste sensation to the anterior two-thirds of the tongue (chorda tympani nerve), preganglionic parasympathetic innervation to the submandibular, sublingual, and *minor salivary glands* (chorda tympani nerve via synapses in the submandibular ganglion), and preganglionic parasympathetic innervation to the nasal mucus glands and lacrimal glands (greater superficial petrosal nerve, via synapses in the pterygopalatine ganglion). (Used with permission from Gilroy AM, MacPherson BR, Ross LM. *Thieme Atlas of Anatomy, Head and Neuroanatomy*. 1st ed. Stuttgart/New York: Thieme;2007, illustration by Karl Wesker.)

and cutaneous blood vessels through the external carotid plexus from the *superior cervical ganglion*. Acetylcholine serves as the neurotransmitter for both postganglionic sympathetic and parasympathetic fibers (**Fig. 6.19**).

The submandibular as well as the sublingual glands are innervated by the secretomotor fibers of the facial nerve (CN VII). Parasympathetic innervation comes from the *superior salivatory nucleus* in the pons, passes through the nervus intermedius and into the internal auditory canal. The fibers are next carried by the *chorda tympani nerve* in the mastoid segment of CN VII, which travels through the middle ear and petrotympanic fissure to the infratemporal fossa. The *lingual nerve*, a branch of the mandibular division of CN V, then carries the presynaptic fibers to the *submandibular ganglion*. The postsynaptic nerve leaves the ganglion to innervate both the submandibular and sublingual glands to secrete serous (watery) saliva. The sympathetic innervation from the *superior cervical ganglion* accompanies the lingual artery to the submandibular tissue and stimulates glandular production of mucoid saliva.

Salivary Gland Secretory Unit

The salivary glands are made up of acini and ducts; together these constitute the secretory unit. The acini contain cells that secrete mucus, serum, or both. These cells drain into the intercalated duct, followed by the striated duct,

and finally into the excretory duct. Myoepithelial cells surround the acini and intercalated duct and serve to expel secretory products into the ductal system. Basal cells along the salivary gland unit replace damaged elements. The combined salivary glands produce 1 to 1.5 L of saliva per day. About 45% is produced by the parotid gland, 45% by the submandibular glands, and 5% each by the sublingual and minor salivary glands (**Table 6.21**).

Table 6.21 Salivary gland secretions

Parotid gland secretion • Proteinaceous, watery, serous secretion • Two-thirds of salivary flow during gustatory and olfactory stimulation • Organic (proteins including enzymes) and inorganic materials are higher.
Submandibular gland secretion • High mucin content, viscous/serous secretion • Higher basal flow rate • Calcium is higher.
Sublingual gland secretion • Higher mucin content than submandibular gland • 5% of salivary flow
Minor salivary gland secretion • Purely mucous glands • 5% of salivary flow

Data from Witt RL, ed. *Salivary Gland Diseases: Medical and Surgical Management*. New York, NY: Thieme;2006.

6.4.1 Salivary Gland Disease

◆ Key Features

- Salivary gland disease may be infectious, inflammatory, noninflammatory, or autoimmune in nature.
- Salivary gland disease may be accompanied by a decrease in salivary flow.
- An underlying autoimmune condition and tumor must be ruled out.

Salivary gland disease may be infectious, inflammatory, noninflammatory, or autoimmune in nature.

◆ Infectious Diseases of the Salivary Glands

Acute Sialadenitis

Acute sialadenitis is an acute inflammation of a salivary gland. Several viral microorganisms have been identified as causative agents for infections

of the salivary glands. The most common virus that causes sialadenitis is the mumps virus (Rubulavirus). Others include coxsackievirus, echovirus, influenza virus, and human immunodeficiency virus (HIV).

Signs and Symptoms

Symptoms of acute sialadenitis include acute painful swelling of the affected salivary gland (Fig. 6.20). At times only one gland is affected initially. There may be swelling of adjacent cervical lymph nodes. The duct orifice may be erythematous. Other symptoms include fever and trismus.

Differential Diagnosis

- Cervical lymphadenopathy
- Acute bacterial sialadenitis
- Inflammatory sialadenitis
- Abscess
- Tumor
- Odontogenic abscess
- Infection

Treatment Options

Treatment is supportive; it typically consists of fluid intake, sialagogues (such as lemon wedges), and analgesics.

Acute Bacterial Sialadenitis

Acute bacterial sialadenitis is a suppurative infection that most commonly infects the parotid glands in debilitated, dehydrated, or elderly patients. Acute bacterial sialadenitis is thought to be an ascending bacterial infection due to decreased salivary flow. Situations that cause a tendency for acute sialadenitis include diabetes mellitus, immunocompromised state, and poor

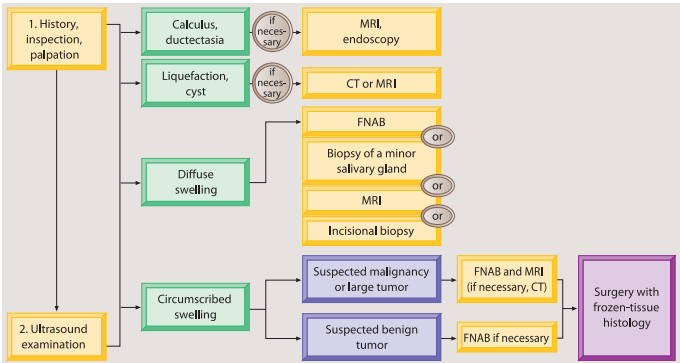


Fig. 6.20 Flowchart for the investigation of salivary gland disease. MRI, magnetic resonance imaging; CT, computed tomography; FNAB, fine-needle aspiration biopsy. (Used with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/New York: Thieme; 2006:141.)

oral hygiene. Other rarer infectious causes of sialadenitis include tuberculosis, actinomycosis, syphilis, and HIV infection.

Signs and Symptoms

Symptoms include diffuse, painful swelling of the affected gland (**Fig. 6.21**). The skin over the gland may be warm, red, and tight. The orifice duct of the affected salivary gland may be red, and massage of the gland may express purulent material from the orifice. Trismus may be present.

Microbiology

The main causative pathogen is *Staphylococcus aureus*. Other bacterial organisms that may cause sialadenitis include *Streptococcus viridans*, *Haemophilus influenzae*, *Streptococcus pyogenes*, and *Escherichia coli*.

Treatment Options

Medical treatment includes antibiotics, hydration, sialagogues (such as lemon wedges), warm compresses, gland massage, and meticulous oral hygiene. If an abscess develops, incision and drainage should be performed with careful attention to the underlying facial nerve. The incision should be made in parallel to the branches of the facial nerve.

◆ **Inflammatory Diseases of the Salivary Glands**

Sjögren's syndrome

Sjögren's syndrome is an inflammatory disease of the exocrine glands. This autoimmune condition may cause chronic sialadenitis. Patients may develop a gradual decrease in salivary production. Xerostomia (dry mouth), xerophthalmia (dry eyes), and lymphocytic infiltration of the exocrine glands is

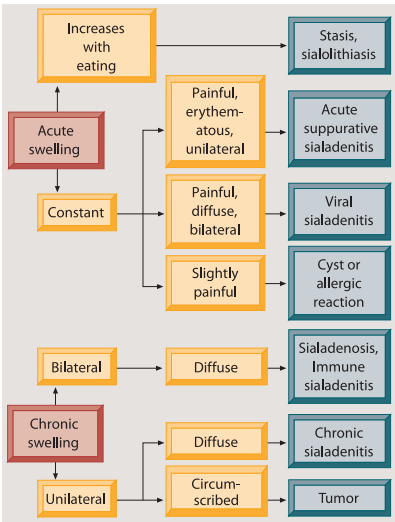


Fig. 6.21 Differential diagnosis of acute and chronic salivary gland swelling. (Used with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/New York: Thieme; 2006:140.)

known as the sicca complex. The pathogenesis is antibodies directed against the antigens of the salivary duct epithelium, causing gland atrophy and interstitial lymphocystic infiltration.

Epidemiology

The disease predominantly affects women.

Signs and Symptoms

Both parotid glands may be involved. Their consistency on palpation is doughy. There may be little or no pain or tenderness. Patients may have associated xerostomia, keratoconjunctivitis, lymphadenopathy, and polyneuropathies.

Labs

An elevated erythrocyte sedimentation rate (ESR) and the presence of auto-antibodies Sjögren's syndrome A (SS-A) and Sjögren's syndrome B (SS-B), rheumatoid factor, and antinuclear antibodies are indicators for the disease.

Other Tests

Biopsy of the minor salivary glands of the lip is diagnostic in 60 to 70% of the cases. Sialography, if performed, may demonstrate a leafless tree pattern.

Treatment Options

Treatment includes immunosuppressant therapy as well as symptomatic therapy with saliva substitutes and artificial tears as well as pilocarpine to augment salivary stimulation.

Complications

Complications of Sjögren's syndrome involving the salivary glands include dental caries. There is an increased incidence of non-Hodgkin lymphomas (NHLs) of the salivary glands among patients with Sjögren's syndrome.

Heerfordt's syndrome

Another cause of chronic sialadenitis is Heerfordt's syndrome, a form of sarcoidosis with enlargement of the parotid glands, mild fever, uveitis, and facial nerve palsy.

◆ Noninflammatory Salivary Disease

Sialolithiasis

Sialolithiasis (salivary stones or calculi) is a stone formation in the excretory duct system of the salivary glands. Sixty to 70% of these are located in the main duct of the gland; 70 to 80% occur in the submandibular gland; 20% occur in the parotid gland. A much smaller percentage may occur in minor salivary glands. Pathogenesis is ductal debris, and calcium phosphate coalesces due to infection, inflammation, or salivary stasis.

Epidemiology

Sialolithiasis typically affects adults; males are more commonly affected than females.

Signs and Symptoms

Symptoms vary according to the degree of obstruction and may include pain and swelling often while or after eating.

Differential Diagnosis

- Inflammation
- Infection
- Tumor

Imaging

Ultrasound may reveal the dilation of the duct system. Seventy to 80% of submandibular salivary stones and only 20% of parotid stones are radiopaque and may be seen on computed tomography (CT) imaging. Noncontrast CT imaging is preferable for detecting salivary stones.

Treatment Options

Typically, the treatment for sialolithiasis is surgical. Intraglandular stones in the submandibular gland typically result in the removal of the gland. Salivary stones that are located near the orifice may be manipulated out. Stones that are more proximal to the gland can be removed either endoscopically (if small enough) or through an incision through the duct system.

Complications

Complications include abscess formation and infection.

Ranula

Ranula is a sialoceles of the sublingual gland typically found in the floor of the mouth. It represents a pseudocyst enlargement. The treatment is surgical excision or marsupialization.

6.4.2 Benign Salivary Gland Tumors

◆ Key Features

- Tumors of the salivary glands represent 2 to 4% of head and neck neoplasms.
- 70% of salivary gland tumors originate in the parotid gland.
- 85% of salivary gland tumors are pleomorphic adenomas (benign mixed tumors).
- Salivary gland tumors most often present as painless masses.
- The most common parotid mass in a child is a hemangioma.

Pleomorphic adenomas are the most common benign salivary gland tumors. They constitute 70% of parotid gland tumors and 50% of

submandibular gland tumors. Pleomorphic adenomas are often located in the tail of the parotid gland. When they are found in the minor salivary glands, the hard palate is the site most frequently involved, followed by the upper lip. Warthin’s tumors (papillary cystadenoma lymphomatousum) occur almost exclusively in the parotid gland. They account for 4 to 11.2% of all salivary gland tumors. These lesions can be multifocal and bilateral (10%) and are more common in individuals who use tobacco (**Table 6.22**).

Table 6.22 Classification of benign primary epithelial salivary gland tumors

Benign mixed tumor (pleomorphic adenoma)
Warthin’s tumor (papillary cystadenoma lymphomatousum)
Oncocytoma (oxyphil adenoma)
Monomorphic tumors
Sebaceous tumors
Benign lymphoepithelial lesion
Papillary ductal adenoma (papilloma)

◆ Clinical

Signs

A painless mass can be felt in the affected salivary gland.

Symptoms

There is slowly increasing, painless swelling of the affected salivary gland. (Pain, rapid growth of long-standing benign parotid mass, and facial nerve paralysis should raise the suspicion of malignant disease; see **Chapter 6.4.3**.)

Differential Diagnosis

- Adenopathy of periparotid or perifacial lymph nodes
- Malignant salivary tumors
- Metastatic lesions
- Autoimmune, infectious, or inflammatory salivary gland lesions

◆ Evaluation

History

History should include questions about time course and onset of mass, pain, relationship of swelling to eating, xerophthalmia, known autoimmune conditions, weight loss, fever, night sweats, and infectious diseases.

Physical Exam

A full head and neck examination, including the salivary glands and the neck, and a bimanual examination of submandibular glands should be performed. Facial nerve function should be documented.

Imaging

Imaging studies are most helpful in the diagnostic evaluation. Ultrasound may help fine-needle aspiration biopsy (FNAB) localization. Magnetic resonance imaging (MRI) is the most sensitive test for establishing the borders of soft tissue tumor extension. Computed tomography (CT) is usually sufficient.

Labs

A white blood cell count should be ordered to investigate for any evidence of leukocytosis and neutrophilic shift, possible infectious process, or lymphoproliferative disease. A lymphoepithelial cyst or recurrent parotid abscess should prompt human immunodeficiency virus (HIV) testing.

Other Tests

FNAB findings provide evidence for a preoperative diagnosis.

Pathology

Pleomorphic adenomas are characterized by variable, diverse, structural histologic patterns. Frequently, they have growth patterns of sheets, strands, or islands of spindle and stellate cells, with a myxoid configuration occasionally predominating. Warthin's tumors are composed of an oncocytic epithelial component that can have a papillary, glandular-cystic, and/or solid growth pattern. Oncocytomas are composed of large oxyphilic cells (oncocytes) (**Fig. 6.22, Fig. 6.23**).

◆ Treatment Options

Medical

Medical therapy is appropriate for infectious and inflammatory processes. Radiotherapy is for nonoperative candidates with pleomorphic adenomas or recurrent multiple pleomorphic adenomas.

Surgical

Management of benign salivary gland tumors includes complete removal with an adequate margin of tissue to avoid recurrences. This involves superficial parotidectomy, total parotidectomy with preservation of facial nerve for deep lobe masses, or submandibular gland removal.

◆ Complications

During submandibular gland excision, unintentional injury may be inflicted on the lingual, hypoglossal, or mandibular (branch of the facial) nerve. Recurrence is usually caused by inadequate excision, tumor enucleation, or tumor spillage.

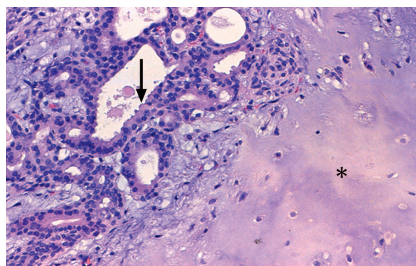


Fig. 6.22 Luminal structures lined with cells showing an apocrine phenotype (arrow) and cartilaginous stroma (asterisk). (Used with permission from Bradley PJ, Guntinas-Lichius O, eds. *Salivary Gland Disorders and Diseases: Diagnosis and Management*. Stuttgart/New York: Thieme;2011:33.)

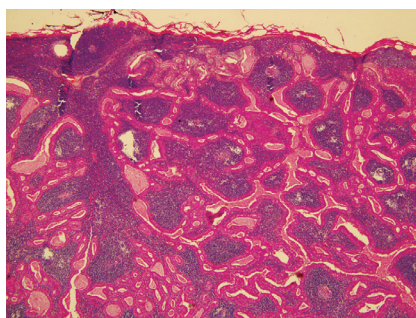


Fig. 6.23 Histology of Warthin tumor. These lesions show bilayered oncocytic epithelium and a dense, lymphoid stroma (hematoxylin/eosin, $\times 40$). (Used with permission from Witt RL, ed. *Salivary Gland Diseases: Surgical and Medical Management*. New York, NY: Thieme; 2006:126.)

With parotidectomy, facial nerve paralysis (paresis) takes a few weeks to resolve spontaneously but can last months or be permanent. Reported rates of permanent postoperative facial nerve paresis range from 0 to 30%. Eye care is important until there is a return of facial nerve function.

Frey's syndrome (gustatory sweating) is caused by an aberrant connection of the postganglionic gustatory parasympathetic fibers to sympathetic fibers of the sweat glands of the overlying skin. It is best avoided by raising a thick flap during parotid surgery.

◆ Outcome and Follow-Up

Evaluate postoperative facial, hypoglossal, and lingual nerve function. Occasionally, transient facial nerve paresis occurs, but it usually resolves within weeks of surgery.

Outcome is typically excellent, and the recurrence rate is very low. Malignant degeneration to carcinoma ex pleomorphic adenoma occurs in 1.4 to 6.3% of untreated pleomorphic adenomas. Malignant degeneration is often associated with a prolonged history of untreated or recurrent pleomorphic adenoma.

6.4.3 Malignant Salivary Gland Tumors

◆ Key Features

- Malignant salivary gland neoplasms account for 0.5% of all malignancies.
- Malignant salivary gland neoplasms account for 3 to 5% of all head and neck cancer.
- The most common malignant major and minor salivary gland tumor is mucoepidermoid carcinoma.
- Numbness or facial nerve weakness and pain in conjunction with salivary gland mass suggest malignancy.

◆ Epidemiology

The frequency of malignant lesions varies by site. Approximately 20 to 25% of parotid tumors, 35 to 40% of submandibular tumors, 50% of palate tumors, and > 90% of sublingual gland tumors are malignant. Mucoepidermoid carcinoma is the most common malignant neoplasm of the salivary glands. Most cases originate in the parotid gland. Mucoepidermoid carcinoma is a malignant epithelial tumor that is composed of various proportions of mucous, epidermoid, intermediate, columnar, and clear cells. Microscopic grading of mucoepidermoid carcinoma is important to determine the prognosis. Mucoepidermoid carcinomas are graded as low, intermediate, and high based on the degree of epidermoid and mucinous cell populations.

Adenoid cystic carcinoma (formerly known as cylindroma) is a slow-growing but aggressive neoplasm with a remarkable capacity for recurrence. This is the most common malignant tumor of the submandibular and minor salivary glands and constitutes 4% of all salivary gland tumors. Morphologically, three growth patterns have been described: cribriform or classic pattern, tubular, and solid or basaloid pattern. This tumor has a propensity for perineural spread. Regardless of histologic grade, adenoid cystic carcinomas, with their unusually slow biologic growth, tend to have a protracted course and ultimately a poor outcome, with a 10-year survival reported to be < 50% for all grades. Many advocate following these patients for the duration of their lifetime, as recurrence can be quite late.

Acinic cell carcinoma is a malignant epithelial neoplasm in which the neoplastic cells express acinar differentiation. In Armed Forces Institute of Pathology (AFIP) data on salivary gland neoplasms, acinic cell carcinoma is the third most common salivary gland epithelial neoplasm; > 80% occur in the parotid gland, women are affected more than men, and the mean patient age is 44 years. Clinically, patients typically present with a slowly enlarging mass in the parotid region. Pain is a symptom in > 33% of patients. For acinic cell carcinoma, staging is likely a better predictor of outcome than histologic grading.

Carcinoma ex pleomorphic adenoma, also known as carcinoma ex mixed tumor or pleomorphic carcinoma, is a carcinoma that shows histologic

evidence of arising from or in a benign pleomorphic adenoma. The neoplasm occurs primarily in the major salivary glands. Diagnosis requires the identification of benign tumor in the tissue sample. The incidence or relative frequency of this tumor varies considerably depending on the study cited.

Lymphomas of the major salivary glands are characteristically of the non-Hodgkin's lymphoma (NHL) type. In an AFIP review of case files, NHL accounted for 16.3% of all malignant tumors that occurred in the major salivary glands, and disease in the parotid gland accounted for ~ 80% of all cases.

Primary squamous cell carcinoma (SCC), also known as primary epidermoid carcinoma, is a malignant epithelial neoplasm of the major salivary glands that is composed of squamous (i.e., epidermoid) cells. Diagnosis requires the exclusion of primary disease located in some other head and neck site; indeed, most SCCs of the major salivary glands represent metastatic disease (Table 6.23).

Table 6.23 Classification of malignant primary epithelial salivary gland tumors

Mucoepidermoid carcinoma
Adenoid cystic carcinoma
Adenocarcinomas ● Acinic cell carcinoma ● Polymorphous low-grade adenocarcinoma ● Adenocarcinoma, NOS ● Rare adenocarcinomas ○ Basal cell adenocarcinoma ○ Clear cell carcinoma ○ Cystadenocarcinoma ○ Sebaceous adenocarcinoma ○ Sebaceous lymphadenocarcinoma ○ Oncocytic carcinoma ○ Salivary duct carcinoma ○ Mucinous adenocarcinoma
Malignant mixed tumors ● Carcinoma ex pleomorphic adenoma ● Carcinosarcoma ● Metastasizing mixed tumor
Rare carcinomas ● Primary squamous cell carcinoma ● Epithelial-myoepithelial carcinoma ● Anaplastic small-cell carcinoma ● Undifferentiated carcinomas ○ Small-cell undifferentiated carcinoma ○ Large-cell undifferentiated carcinoma ○ Lymphoepithelial carcinoma ● Myoepithelial carcinoma ● Adenosquamous carcinoma

Abbreviation: NOS, not otherwise specified.

Malignant neoplasms whose origins lie outside the salivary glands may involve the major salivary glands by:

1. Direct invasion from cancers that lie adjacent to the salivary glands
2. Hematogenous metastases from distant primary tumors
3. Lymphatic metastases to lymph nodes within the salivary gland

Direct invasion of nonsalivary gland tumors into the major salivary glands is principally from squamous cell and basal cell carcinomas of the overlying skin. Prior exposure to ionizing radiation appears to substantially increase the risk for development of malignant neoplasms of the major salivary glands.

◆ Clinical

Signs and Symptoms

A painless swelling of the affected salivary gland is a sign of a salivary gland tumor. Approximately 10 to 15% of malignant parotid neoplasms present with pain. Occasionally, malignant salivary gland tumors may be characterized by rapid growth or a sudden growth spurt. Numbness or nerve weakness caused by nerve involvement and persistent facial pain are highly suggestive of malignancy. Depending on the site of the primary tumor, other symptoms include drainage from the ipsilateral ear, dysphagia, trismus, and facial paralysis. The overlying skin or mucosa may become ulcerated.

Differential Diagnosis

- Adenopathy of periparotid or perifacial lymph nodes
- Benign salivary tumors
- Metastatic lesions
- Autoimmune, infectious, or inflammatory salivary gland lesions

◆ Evaluation

History

History includes questions about time course and onset of mass, pain, facial weakness, weight loss, known autoimmune disease, fever, and night sweats.

Physical Exam

A full head and neck examination including salivary glands, neck, and a bimanual examination of submandibular glands should be performed. Cranial nerve (CN) function should be assessed.

Imaging

Imaging studies are helpful in the diagnostic evaluation. Ultrasound may help fine-needle aspiration biopsy (FNAB) localization. Magnetic resonance imaging (MRI) is the most sensitive test for establishing the borders of soft tissue tumor extension. Contrast-enhanced computed tomography (CT) is usually sufficient.

Other Tests

FNAB findings often provide evidence for a preoperative diagnosis.

Pathology

Salivary gland neoplasms show extreme histologic diversity. These neoplasms include malignant tumors of epithelial, mesenchymal, and lymphoid origin. There are several characteristic appearances that are notable. Mucoepidermoid carcinoma, as the name suggests, will have a combination of mucinous cells and squamous cells that become more de-differentiated in higher-grade tumors. Adenoid cystic carcinoma, especially the cribriform subtype, can have a “Swiss cheese” appearance on histology. Histologic grading (Table 6.24) of salivary gland carcinomas is important to determine the proper treatment approach. However, it is not an independent indicator of the clinical course and must be considered in the context of the clinical stage; see Staging of Malignant Salivary Gland Tumors (Fig. 6.24).

Table 6.24 Histologic grading of salivary gland carcinomas

Low-grade • Acinic cell carcinoma • Basal cell adenocarcinoma • Clear cell carcinoma • Cystadenocarcinoma • Epithelial-myoepithelial carcinoma • Mucinous adenocarcinoma • Polymorphous low-grade adenocarcinoma
Low-, intermediate, and high-grade • Adenocarcinoma, NOS • Mucoepidermoid carcinoma • Squamous cell carcinoma
Intermediate and high-grade • Myoepithelial carcinoma
High-grade • Anaplastic small-cell carcinoma • Carcinosarcoma • Large-cell undifferentiated carcinoma • Small-cell undifferentiated carcinoma • Salivary duct carcinoma

Abbreviation: NOS, not otherwise specified.

◆ Treatment Options

Medical

The use of chemotherapy for malignant salivary gland tumors remains under evaluation. Postoperative radiotherapy augments surgical resection, particularly for the high-grade neoplasms, when margins are close or

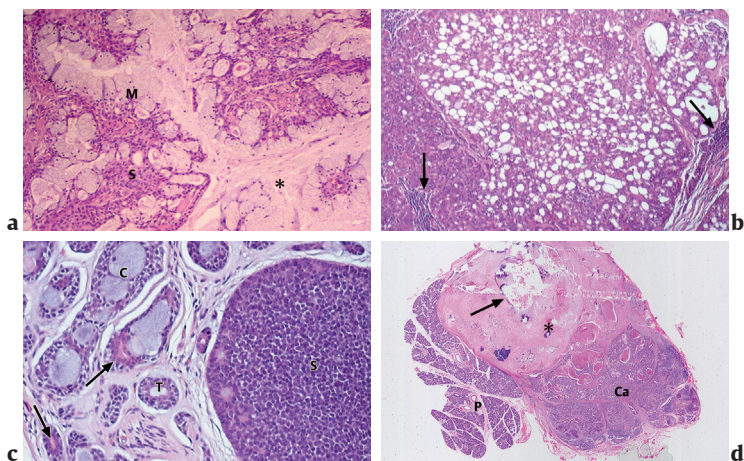


Fig. 6.24 (a) Slightly hematoxyphilic mucous cells (M), eosinophilic squamous cells (S), and stromal pooling of mucosubstances (asterisk). (b) A microcystic acinic cell carcinoma (ACC) with stromal lymphoid aggregates (arrows). (c) Solid (S), cribriform (C), and tubular (T) arrangements, showing lumina lined with eosinophilic cells (arrows). (d) The “ghost” of a pleomorphic salivary adenoma (asterisk) with fragmented calcification (arrow). Ca, a malignant invasive component; P, parotid. (Used with permission from Bradley PJ, Guntinas-Lichius O, eds. *Salivary Gland Disorders and Diseases: Diagnosis and Management*. Stuttgart/New York: Thieme; 2011:36–38.)

involved, when tumors are large, or when histologic evidence of lymph node metastases is present. Fast neutron-beam radiation or accelerated hyperfractionated photon beam schedules have been reported to be more effective than conventional X-ray therapy in the treatment of inoperable, unresectable, or recurrent malignant salivary gland tumors.

Surgical

The minimum therapy for low-grade (and intermediate-grade mucoepidermoid carcinoma) malignancies of the superficial portion of the parotid gland is a superficial parotidectomy. For all other lesions, a total parotidectomy with facial nerve preservation is often indicated. The facial nerve or its branches should be resected if involved by tumor. Rarely, a mastoidectomy or temporal bone resection may be performed to obtain a negative margin on CN VII. Neck dissection may be performed, depending on tumor type and nodal involvement.

◆ Complications

Parotidectomy

Recurrence may be caused by inadequate excision, positive margins, or tumor spillage. Facial nerve paralysis and cosmetic deformity may need to be addressed surgically. CN dysfunction from the tumor and resultant surgery is a possible complication. Eye care is important in the presence of facial nerve paralysis.

◆ Outcome and Follow-Up

Early-stage low-grade malignant salivary gland tumors are usually curable by adequate surgical resection alone. The prognosis is more favorable when the tumor is in a major salivary gland. Large, bulky tumors or high-grade tumors carry a poorer prognosis. The prognosis also depends on the following: the gland in which they arise, histology, the grade, the stage, perineural involvement, and spread to adjacent structures, lymph nodes, or distant sites. The prognosis for any treated cancer patient with progressing or relapsing disease is poor, regardless of histologic type or stage. It is notable that adenoid cystic carcinoma can recur in a delayed fashion, with some recurrences occurring more than 10 years after treatment of the primary tumor.

◆ Staging of Malignant Salivary Gland Tumors

Primary Tumor (T)

- TX: Primary tumor cannot be assessed
- T0: No evidence of primary tumor
- Tis: Carcinoma in situ
- T1: Tumor ≤ 2 cm in greatest dimension without extraparenchymal extension
- T2: Tumor > 2 cm and ≤ 4 cm in greatest dimension without extraparenchymal extension
- T3: Tumor > 4 cm and/or tumor having extraparenchymal extension
- T4:
 - T4a: Moderately advanced disease; tumor invades the skin, the mandible, the ear canal, and/or the facial nerve
 - T4b: Very advanced disease; tumor invades the skull base and/or the pterygoid plates and/or encases the carotid artery

*Extraparenchymal extension is clinical or macroscopic evidence of invasion of soft tissues. Microscopic evidence alone is not sufficient.

Regional Lymph Nodes (N)*

- NX: Regional lymph nodes cannot be assessed
- N0: No regional lymph node metastasis
- N1: Metastasis to a single ipsilateral lymph node measuring ≤ 3 cm in greatest diameter and ENE negative
- N2: Further divided into three categories:
 - N2a: Single ipsilateral lymph node > 3 and ≤ 6 cm and ENE negative (pathologic staging also includes a single ipsilateral or contralateral lymph node ≤ 3 cm and ENE positive)
 - N2b: Multiple ipsilateral lymph nodes ≤ 6 cm and ENE negative
 - N2c: Bilateral or contralateral lymph nodes ≤ 6 cm in greatest dimension and ENE negative
- N3: Now further divided into two categories:
 - N3a: Metastasis to a lymph node > 6 cm, ENE negative

A designation of “U” or “L” can be attached to the N category to indicate nodes above (“U”) or below (“L”) the lower border of the cricoid. Similarly, clinical and pathologic ENE should be recorded as ENE (–) or ENE (+). Clinical and pathologic staging are similar with the differences noted in parenthesis.

Distant Metastasis (M)

M1: Distant metastasis

Stage 0 disease: Carcinoma in situ (TisNOM0)

Stage I disease: Includes only T1 N0 M0 tumors

Stage I disease: Includes only T1 N0 M0 tumors

Stage II disease: Includes only T2 N0 M0 tumors

Stage III disease: Includes T3 N0 M0 and T1 -3 disease that is N1 M0

Stage IVA disease: Includes T4a disease with N0-N2 M0 disease and T1-T3 disease that is N2 M0

Stage IVB disease: Includes T4b disease with any nodal disease or T1-T4a disease with N3 disease without distant metastases (M0)

Stage IVC disease: Any disease with distant metastases (M1)

See **Table 6.7** in Chapter 6.3.3.